

PREDICTIVE RESEARCH LAB OS

# Lucky Loop

Predict before compute. Verify before claim.

A research-agent backend where Qwen-AgentWorld forecasts every expensive lab action, real Python tests reality, and a deterministic verifier decides what claims survive.

Qwen-AgentWorld

Real sklearn execution

Claim ledger

Audit-ready UI

The screenshot displays the CURZE ORACLE interface, which is a predictive research lab OS. The top header shows the model name 'Qwen/Qwen-AgentWorld-35B-A3B' and the mode 'PREVIOUS RUN'. The main content area is divided into several sections:

- CURRENT STATE:** Displays the state ID 'OPEN-CAN-A', timestamp 'Sun, 28 Jun 2026 07:55:19 GMT', and context 'Open auto-research run completed.' It also shows a horizon of 8 and a temperature of 0.20.
- GOONS:** A list of actions with their state IDs, probabilities, and anomalies. The actions are: 'ACTION\_008\_STOP\_...' (prob. 0.50, anomaly), 'ACTION\_007\_RUN\_P...' (prob. 0.50, anomaly), and 'ACTION\_006\_DRY\_R...' (prob. 0.50, anomaly).
- VERDICT (TOP OUTCOME):** A red box indicating 'CLAIM BLOCKED.' with the message 'The verifier rejected the robust superiority claim. Baseline logistic regression stands.'
- PAPERS:** A list of research papers, including 'Evaluating Machine Learning Models for Supernova Gravitational Wave Signal Classification' and 'Benchmarking Logistic Regression, SVM, and LightGBM Against BiLSTM with Attention for Sentiment Analysis on Indonesian Product Reviews'.
- CLEAN TRACES:** A table showing the execution of various actions, including 'Open auto-research run completed', 'Lab action completed', 'Executing lab action', and 'Qwen-AgentWorld prediction received'.
- STATE DIFFS:** A table showing the differences in state between different runs, including transitions like 'S\_7C1A9F28 -> S\_7C1AA3D1' and 'S\_7C1A9F28 -> S\_7C1AB7E9'.

The interface also includes a sidebar with navigation options like 'QUERY', 'GOONS', 'PAPERS', 'TRACES', and 'STATES'. At the bottom, there is a footer with user information 'USER: NIGHT HAUNTER', clearance level 'CLEARANCE: DREAD', and query details 'QUERY ID: Q\_CAN-A-R0', 'TOKENS: 0', and 'CONTEXT: 128K'.

## THE PROBLEM

# Most AI research agents learn the story after the experiment.

### They waste compute

Actions are launched before the system estimates runtime, failure mode, value of information or protocol risk.

### They overclaim metrics

A single best score can become a confident scientific claim even when seed noise or leakage risk dominates.

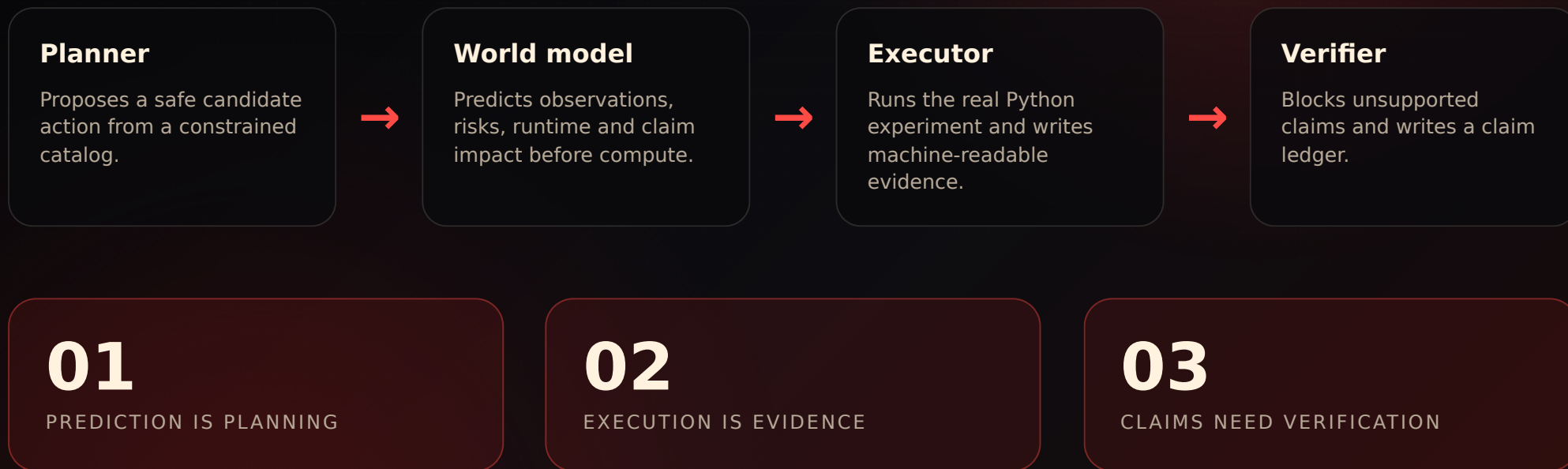
### They hide misses

Reports rarely preserve what the model predicted before execution, so failures disappear from the narrative.

The missing primitive is not another experiment runner. It is a pre-compute world model plus a claim gate.

## THE INSIGHT

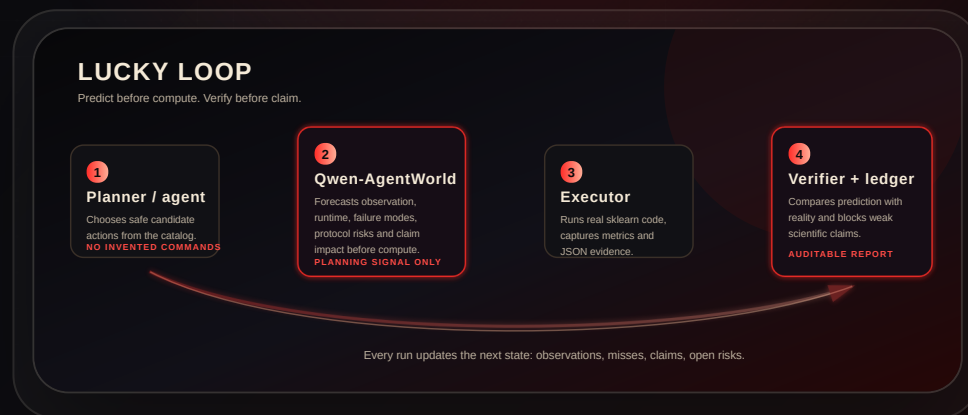
# Separate prediction, execution and scientific claims.



## PRODUCT

# An autonomous ML lab pipeline from question to honest report.

- Starts from a natural-language research question.
- Builds domain and method literature context.
- Searches Hugging Face/OpenML datasets and audits candidates.
- Generates a protocol, validates code, dry-runs, then executes.
- Writes events, notebook, predictions, analyses, ledger and final report.





DEMO SURFACE

# The cockpit makes the research trace visible.

**CURZE ORACLE**  
WORLD MODEL · PREDICT NEXT STATES

MODEL  
Qwen/Qwen-AgentWorld-35B-A3B

MODE  
PREVIOUS RUN

STATUS  
ARCHIVED

QUERY

GOONS

PAPERS

TRACES

STATES

VERDICT

**CURRENT STATE**

State ID  
OPEN-CAN-A

Timestamp (UTC)  
Sun, 28 Jun 2026  
07:55:19 GMT

Context  
Open auto-research run  
completed.

HORIZON  
8

TEMPERATURE  
0.20



1  
STATE ID  
ACTION\_008\_STOP\_  
PROB.  
0.50  
ANOMALY stop and report

2  
STATE ID  
ACTION\_007\_RUN\_P  
PROB.  
0.50  
ANOMALY run protocol

3  
STATE ID  
ACTION\_006\_DRY\_R  
PROB.  
0.50  
ANOMALY dry run

VERDICT (TOP OUTCOME)

**CLAIM BLOCKED.**

The verifier rejected the robust superiority claim. Baseline logistic regression stands.

Ask Curze...

Turns 8

PAPERSVIEW ALL

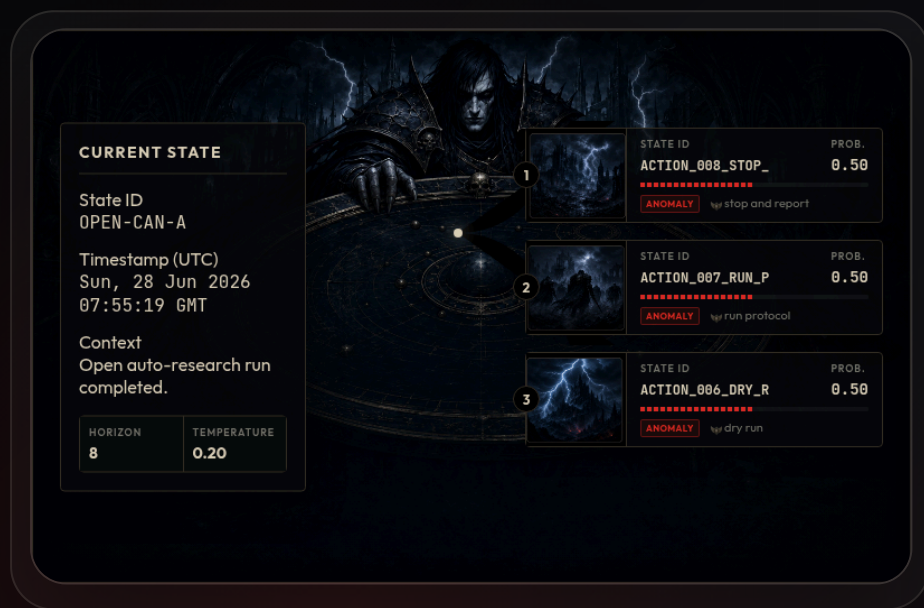
CLEAN TRACESVIEW ALL

STATE DIFFSVIEW ALL

## What the UI shows

- Current lab state
- Predicted branches
- Live pipeline events
- Literature sources
- Clean traces and verdict
- Downloadable report

# Qwen-AgentWorld predicts the next lab state before the action runs.



**CURRENT STATE**

State ID  
OPEN-CAN-A

Timestamp (UTC)  
Sun, 28 Jun 2026  
07:55:19 GMT

Context  
Open auto-research run  
completed.

|         |             |
|---------|-------------|
| HORIZON | TEMPERATURE |
| 8       | 0.20        |

**Prediction contract**

1  
STATE ID  
ACTION\_008\_STOP\_  
PROB.  
0.50  
ANOMALY stop and report

2  
STATE ID  
ACTION\_007\_RUN\_P  
PROB.  
0.50  
ANOMALY run protocol

3  
STATE ID  
ACTION\_006\_DRY\_R  
PROB.  
0.50  
ANOMALY dry run

## Prediction contract

expected observation  
expected artifacts  
runtime / compute-waste risk  
failure modes  
protocol risks  
claim support probability  
recommendation: run | verify | modify | skip | stop

The prediction does not count as evidence. It is a planning signal that must later be compared with reality.

# The verifier creates the honest moment.

## What gets blocked

Apparent winner:  $C=0.1$   
effect\_size = 0.020979  
seed\_noise = 0.027972  
verdict = inconclusive  
blocked: robustly better  
allowed: best mean, but effect smaller than seed noise

## claim ledger

EVERY VERDICT IS PRESERVED

### Blocked claim

Strong wording is rejected when evidence is insufficient.

### Allowed rewrite

The report can still say what the evidence genuinely supports.

# Every run becomes an audit folder, not just a pretty answer.

## Generated artifacts

```
reports/lab/<slug>/  
literature/*/sources.json  
agenda/research_agenda.json  
datasets/selected_dataset.csv  
protocol/generated_protocol.json  
generated/experiment.py  
predictions/world_model_predictions.jsonl  
notebook.jsonl  
events.jsonl  
claim_ledger.json  
final_report.md
```

| PAPERS                                                                                                                                                                                                                                   |                                           | VIEW ALL ▾                     |                                    |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------|------------------------------------|-------|
| <div>Evaluating Machine Learning Models for Supernova Gravitational Wave Signal Classification</div> <div>Y. Sultan Akyltasov et al. (2024) <a href="#">2409.14508</a></div> <div>TXT</div>                                              |                                           |                                |                                    |       |
| <div>Benchmarking Logistic Regression, SVM, and LightGBM Against BiLSTM with Attention for Sentiment Analysis on Indonesian Product Reviews</div> <div>Y. Sultan Akyltasov et al. (2024) <a href="#">2409.14508</a></div> <div>TXT</div> |                                           |                                |                                    |       |
| <div>arXiv:2409.14508v1 [astro-ph] 10 Sep 2024</div>                                                                                                                                                                                     |                                           |                                |                                    |       |
| CLEAN TRACES                                                                                                                                                                                                                             |                                           | VIEW ALL ▾                     |                                    |       |
| TIME (UTC)                                                                                                                                                                                                                               | SOURCEEVENT                               | STATUS                         |                                    |       |
| 09:55:19                                                                                                                                                                                                                                 | CLEAN Open auto-research run completed.   | CLEAN                          |                                    |       |
| 09:55:19                                                                                                                                                                                                                                 | CLEAN Lab action completed.               | CLEAN                          |                                    |       |
| 09:55:19                                                                                                                                                                                                                                 | CLEAN Executing lab action.               | CLEAN                          |                                    |       |
| 09:55:19                                                                                                                                                                                                                                 | FEED Qwen-AgentWorld prediction received. | CLEAN                          |                                    |       |
| STATE DIFFS                                                                                                                                                                                                                              |                                           | VIEW ALL ▾                     |                                    |       |
| TRANSITION                                                                                                                                                                                                                               | BEFORE                                    | AFTER                          | DELTA                              | CONF. |
| S_7C1A9F28 → S_7C1AA3D1                                                                                                                                                                                                                  | Stability: 0.71<br>Order: 0.65            | Stability: 0.17<br>Order: 0.21 | + Stability -0.54<br>+ Order -0.44 | 0.62  |
| S_7C1A9F28 → S_7C1AB7E9                                                                                                                                                                                                                  | Stability: 0.71<br>Order: 0.65            | Stability: 0.38<br>Order: 0.42 | + Stability -0.33<br>Order -0.23   | 0.24  |
| S_7C1A9F28 → S_7C1AC4F6                                                                                                                                                                                                                  | Stability: 0.71<br>Order: 0.65            | Stability: 0.29<br>Order: 0.31 | + Stability -0.42<br>Order -0.34   | 0.14  |



DIFFERENTIATION

# Why this is different from a classic auto-research loop.

| Capability                  | Classic research agent       | Lucky Loop                                           |
|-----------------------------|------------------------------|------------------------------------------------------|
| Pre-compute forecast        | No — actions run first       | Yes — Qwen-AgentWorld predicts observations and risk |
| Prediction-vs-reality trace | Usually absent               | Logged as first-class evidence                       |
| Claim discipline            | Often narrative-based        | Deterministic verifier and claim ledger              |
| Generated-code safety       | Variable                     | Static validation plus dry-run before full compute   |
| Demo honesty                | Best score becomes the story | Blocked claims remain visible                        |

## DEPLOYMENT

# Built as a backend, UI and API surface that can swap agents.

### Product path

Scientist planner through any OpenAI-compatible endpoint. World model through Qwen-AgentWorld. Python owns execution, verifier and reports.

### Agent-in-repo path

Codex, Claude Code, OpenClaw or Hermes can operate from repo runbooks while the backend keeps trace labels explicit.

### Development path

No-model smoke mode validates local code, artifacts and reports without pretending a model was called.

```
PYTHONPATH=src python web/server.py 8000  
PYTHONPATH=src python -m luckyloop.lab --question "..." --budget 8  
PYTHONPATH=src python scripts/validate_lab_artifacts.py --workspace reports/lab/<slug>
```

# From impressive demo to credible open research infrastructure.

01

## Freeze demo loop

Stable run, UI, report export and reproducibility evidence.

02

## Ablation proof

Compare classic, classic+verifier and full Lucky Loop policies.

03

## Agent adapters

Cleaner OpenAI-compatible planner and agent-in-repo modes.

04

## Research packs

Curated protocols beyond sklearn tabular classification.

The immediate win is a 60-second story: forecast  
→ execute → compare → block or allow the claim.

CLOSE

# Most AI scientists hallucinate after the experiment.

Lucky Loop makes a prediction before the experiment, runs the real code, compares prediction with reality, and only claims what survives verification.

**predict**

BEFORE COMPUTE

**execute**

REAL CODE

**verify**

BEFORE CLAIM

Lucky Loop

Open auto-research backend

Audit-ready scientific claims